The G_ε^μ-lemma, reduced

Fejér on the occupied cosine grid, the digamma multiplier, and why the box is not construction-pure

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Abstract

Round 381 reduced G_ε to the triple $G_\varepsilon^\mu \wedge C_\varepsilon \wedge R_2$. Rounds 385–389 closed the signed half as a cancellation object. This note attacks the remaining positive sibling: the last twelve monic Jacobi coefficients of the μ-chain satisfy $|\gamma_k - \frac{1}{4}| \le \frac{1}{16}$ and $|\log(\gamma_{k+1}/\gamma_k)| \le \frac{2}{5}$. What is proved: the Jacobi-$(0,1)$ identity (r379); Chebyshev-U quarters $\gamma_k = \frac{1}{4}$; homogeneity of monic γ; and that Fejér weights on the *full* cosine grid are a discrete Bernstein–Szegő measure ($\max_{k\ge2}|\gamma_k - \frac{1}{4}| = 1.8 \cdot 10^{-6}$ on FRAME-A $w = 9$). What is *not* proved: the box. Occupied-only Fejér (the ν-complement deleted) already lies in $|\gamma - \frac{1}{4}|$ (core-42 max 0.04892), with jump margin 0.0058 against $\frac{2}{5}$. The d_{arm} multiplier pulls *in* on 40/42 and is load-bearing for jump headroom. A ratio bound $\kappa = \sup(1+\varepsilon)/\inf(1+\varepsilon) = 4200$ is true and useless. A permutation of the same positive weights on the same nodes leaves the box on 17/20 seeds: G_ε^μ is not construction-pure. The hydra-legal remainder is the pair $(F_\varepsilon, W_\varepsilon)$: Fejér-on-occupied (node deletion from Bernstein–Szegő), and the r342 digamma/tent multiplier versus scramble-of-weights. No RH claim.

Status. Lemmas 1–3 are proved (finite identities). Lemma 4 is a machine identity on the full cosine grid. Lemma 6 is a true majorant that does not yield the box. Proposition 1 is a finite machine census, not a cofinal law. Proposition 2 names the reduction. No RH claim. No G_ε claim.

1 Recollection

G_ε^μ is the positive sibling of G_ε in [1]: writing $n = N - 2$ for the v_2-depth, the last twelve monic Jacobi coefficients of the μ-chain (occupied cosine nodes, weights Fejér $\times d_{\mathrm{arm}}$) satisfy

$$\left|\gamma_k - \tfrac{1}{4}\right| \le \tfrac{1}{16}, \qquad \left|\log(\gamma_{k+1}/\gamma_k)\right| \le \tfrac{2}{5}, \qquad k \in [n-12, n). \tag{1}$$

The mesh $x = \cos(2\pi k/L)$, $L = 4h - 2$, is a construction identity of [2]. The source weights follow the r342 dictionary $F_A(\xi) = -\log \pi + \Re\psi(\frac{1}{4} + i\xi/2)$ plus the two-cell tent of the prime layer, $v_{\mathrm{pred}} < 10^{-4}$. There is no signed cancellation on this chain: the four dead majorant routes of the evening do not apply. The r381 measurement on FRAME-A $w = 9$ is last-twelve 0.02605, jump 0.1224.

2 The Fejér basis

Lemma 1 (Jacobi-$(0,1)$)**.** *For the continuous weight* $(1-x)$ *on* $[-1,1]$,

$$\gamma_n = \frac{n(n+1)}{(2n+1)^2}, \qquad \left|\gamma_n - \tfrac{1}{4}\right| = \frac{1}{4(2n+1)^2}.$$

Exact over $\mathbb{Q}$ *for* $n = 1, \dots, 79$.

This is the r379 identity. It is *not* the discrete Fejér chain on the cosine grid: that grid with weights $1-x$ samples a Bernstein–Szegő measure (Chebyshev-U class), not Gauss–Jacobi nodes of $(0,1)$.

Lemma 2 (Chebyshev-U quarters)**.** *The monic Chebyshev polynomials of the second kind satisfy* $p_2 = x^2 - \frac{1}{4}$ *and* $\gamma_k = \frac{1}{4}$ *for every* $k \geq 1$*.*

Proof. $U_0 = 1$, $U_1 = 2x$, $U_2 = 4x^2 - 1$. The monic forms are $p_0 = 1$, $p_1 = x$, $p_2 = x^2 - \frac{1}{4} = x\,p_1 - \frac{1}{4}\,p_0$. □

Lemma 3 (homogeneity)**.** *Monic Jacobi coefficients are invariant under* $w \mapsto c\,w$ *for* $c > 0$*. Exact over* $\mathbb{Q}$ *on a five-atom toy.*

Lemma 4 (full-grid Fejér)**.** *On the unique cosine folds* $x = \cos(2\pi k/L)$*,* $k = 1, \ldots, L/2$*, with Fejér weights* $1-x$*, the monic Stieltjes coefficients of FRAME-A* $w = 9$ *(*$S = 367 = S_+ + S_-$*, depth* 182*) satisfy* $\max_{k\geq 2}|\gamma_k - \frac{1}{4}| = 1.84 \cdot 10^{-6}$*. The union of* μ *and* ν *atoms* is *this grid.*

Occupied-only Fejér is therefore a *node deletion*: the ν-complement is removed from a Bernstein–Szegő discrete measure. The r381 figure 0.071 is signed Fejér ($\mu - \nu$ with Fejér weights on both sides) and lives in C_ε, not here.

Lemma 5 (jump not implied)**.** $\log\frac{5}{3} > \frac{2}{5}$*. The jump half of* (1) *is not a corollary of the* $|\gamma - \frac{1}{4}|$ *box.*

3 Positive perturbation

Write $w = w_{\text{Fejer}} \cdot (1+\varepsilon)$ on the occupied nodes, with ε the Fejér-stripped d_{arm} (r342 profile).

Lemma 6 (ratio bound, useless)**.** *If* $m \leq 1 + \varepsilon \leq M$ *pointwise then* $(m/M)\gamma_k(w_{\text{Fejer}}) \leq \gamma_k(w) \leq (M/m)\gamma_k(w_{\text{Fejer}})$*. On FRAME-A* $w = 9$*,* $\kappa = M/m = 4200.6$ *(*$\|\varepsilon\|_\infty = 5.166$*). The image of the occupied-Fejér last twelve under this bound is* $[10^{-4}, 1163]$*, which misses* $[\frac{3}{16}, \frac{5}{16}]$*.*

The bound is legitimate — everything is positive — and too crude. Last-twelve-relevant mass (nodes carrying 90% of $p_{n-6}^2 w$) still sees $\kappa = 4200$. Nevai–Blumenthal / Rakhmanov stability under a relative weight bound on a fixed infinite support is an $o(1)$ statement as $k \to \infty$; here $k \sim n = N-2$ is half-filling depth on a *finite* atomic subset, and no rate yields $\frac{1}{16}$.

Directed measurement, not a bound: at $w = 9$, $|\gamma(\mu) - \gamma(\text{Fejer})|_{12} = 0.030$ and $|\gamma(\mu) - \frac{1}{4}|_{12} = 0.026 < 0.036 = |\gamma(\text{Fejer}) - \frac{1}{4}|_{12}$. The d_{arm} multiplier pulls *in*.

4 Census and cofinality

Proposition 1 (core-42 census)**.** *On the* 42 *admissible MAIN windows, last-twelve of the* μ*-chain and of occupied-Fejér both satisfy* (1)*:* μ *max* 0.03304 *(kz* 14*), max jump* 0.2469*; occupied-Fejér max* 0.04892 *(kz* 55*), max jump* 0.3942*. The* d_{arm} *multiplier pulls toward* $\frac{1}{4}$ *on* 40/42*. Builder fallback: core-*42 *only (MAIN-*85 *aborted above two minutes on EXT windows).*

There is no m_4. The smallest core window ($n = 140$) already lies in the box; depth improves the numbers (kz 82, $n = 532$: 0.0085) but is not a threshold. The box on this family is a sealed census, not a cofinal theorem.

The jump half of occupied-Fejér is razor-thin (margin 0.0058 against $\frac{2}{5}$). That is where d_{arm} is load-bearing: it supplies jump headroom the Fejér-on-occupied chain barely has.

5 Adversaries

Permutation on the same nodes. Twenty independent shuffles of the $w = 9$ μ-weights on the fixed occupied cosine nodes: 17/20 leave (1) (worst last-twelve 0.101, jump 0.621). Seed 3 is the named kill (0.07445, jump 0.459). Seed 1 happened to stay in — a single permutation is not a theorem of positivity plus the grid.

Scramble-seed μ. Window scramble seed 1 (occupation *and* weights move): μ-only last-twelve 0.07407, jump 0.3518, out of the box. The node set selected by $\operatorname{sign}(d_{\text{arm}})$ is part of the construction; scrambling it is a different support.

Two-period weight. On occupied Fejér, the multiplier $1 + a(-1)^i$ with $0 \le a \le 0.95$ stays in the box ($a = 0.95$: last-twelve 0.0497, jump 0.385). Smooth oscillation of the weight is not the kill; spatial scramble of the d_{arm} multiset is.

Mutants. The $|\gamma - \frac{1}{4}|$ half at $w = 9$ still holds against $\frac{1}{32}$ ($0.02605 < 0.03125$). The jump half fails against $\frac{1}{10}$ ($0.1224 > 0.10$). Tightening the jump bar to 0.39 would fail occupied-Fejér on kz 55.

Equal weights, power weights. Equal weights on the occupied nodes, and weights $(1-x)^p$ for $p \in \{-0.4, 0, 0.5, 1, 2, 4\}$, all stay in the box at $w = 9$. Regular profiles compatible with Fejér do not kill; the r342 assignment of which mass sits on which fold does.

6 What remains

Proposition 2 (reduction)**.** *G^μ_ε reduces to the pair $(F_\varepsilon, W_\varepsilon)$:*

- *F_ε: last-twelve box* (1) *for Fejér weights on the occupied cosine subset (node deletion of the ν-complement from Lemma 4; equivalently the F1-bounded ν-runs of [3] acting on Bernstein–Szegő discrete).*
- *W_ε: the r342 digamma/tent multiplier does not kick that chain out of* (1) *(permutation of the same masses on the same nodes is a named counterexample to any claim that positivity plus the grid suffice).*

Both are strictly smaller than G^μ_ε: F_ε freezes the weights, W_ε is a multiplicative perturbation of a named reference. G^μ_ε is not construction-pure. The arithmetic content of the G_ε triple that lives in the weights *(not the nodes) is W_ε here and C_ε on the signed side. The chain*

$$T_2' \Longleftarrow \cdots \Longleftarrow G_\varepsilon \Longleftarrow G^\mu_\varepsilon \wedge C_\varepsilon \wedge R_2 \Longleftarrow F_\varepsilon \wedge W_\varepsilon \wedge C_\varepsilon \wedge R_2$$

stands. No m_4. No RH claim.

A Geronimus/Szegő bound that would absorb W_ε into F_ε is exactly the ratio majorant of Lemma 6, and it does not close. T1 / COMPOSE unchanged.

7 Machine checks

Numbered lemmas: `verify_g_eps_mu.py` (same directory). Standalone: Jacobi-$(0,1)$; Chebyshev-U; homogeneity; jump comparison; small full-grid Fejér; ratio-bound toy. Construction: $w = 9$ μ / occupied-Fejér / signed-Fejér, full-grid residual, κ, permutation kill, scramble-seed μ, two-period weight. Discovery census: `g_eps_mu_probe.py` (`--smoke` / full), core-42. Exit line: `G_EPS_MU LEMMA VERIFIED`.

Claim boundary

Finite identities for Jacobi coefficients of positive atomic measures on a cosine grid, a named reduction of G_ε^μ to $(F_\varepsilon, W_\varepsilon)$, and named permutation / scramble breaks. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

[1] S. Hamann, *The G_ε-lemma, reduced*, standalone note, August 2026, `rh/problem/g_eps_lemma.tex`.

[2] S. Hamann, *The V_3'-lemma, reduced*, standalone note, August 2026, `rh/problem/v3prime_proof.tex`.

[3] S. Hamann, *The pivot-band entry lemma, reduced*, standalone note, August 2026, `rh/problem/pivot_entry_lemma.tex`.